lab2

Problem 1:

• Run a container nginx with name my-nginx and attach 2 volumes to the container using volume mount

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name my-nginx -v html_volume:/
usr/share/nginx/html -v nginx_config_volume:/etc/nginx nginx
41d482f38d6e16a294eb2f243f0d45d5c5fdee56f3b2a6951466350189bdf300
```

Volume1 for containing static html file

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker volume create html_volume
html_volume
```

Volume2 for containing nginx configuration

```
html_volume
albraa@albraa-Cyborg-15-A13UDX:~$ docker volume create nginx_config_volume
nginx_config_volume
```

Edit the html content

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker exec -it my-nginx bash
root@41d482f38d6e:/# cd /usr/share/nginx/html
root@41d482f38d6e:/usr/share/nginx/html# echo "<h1> hello albraa from docker Ngi
nx </h1>" > index.html
root@41d482f38d6e:/usr/share/nginx/html# exit
exit
```

Remove the container

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps -a
CONTAINER ID   IMAGE                 COMMAND                  CREATED         STAT
US                    PORTS     NAMES
41d482f38d6e   nginx                 "/docker-entrypoint.…"   2 minutes ago   Up 2
 minutes              80/tcp    my-nginx
e08fd09ddd58   simple-python-app     "python3 app.py"         4 hours ago     Exit
ed (0) 4 hours ago             nervous_leavitt
afd2554cd7a9   0236ee02dcbc          "/docker-entrypoint.…"   4 hours ago     Exit
ed (0) 3 hours ago             webs
b49c7045b4fb   mysql:latest          "docker-entrypoint.s…"   4 hours ago     Exit
ed (0) 3 hours ago             app-database
albraa@albraa-Cyborg-15-A13UDX:~$ docker rm -f my-nginx
my-nginx
```

Run a new 2 containers with the following:

o Attach the two volumes that were attached to the previous container

using volume mount

o Map port 80 to port 8080 on you host machine

o Access the html files from your browser

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name nginx2 -v html_volume:/us
r/share/nginx/html -v nginx_config_volume:/etc/nginx nginx
ea60b2cb873d45b03a5d528495f06afe37d004a5f82f22c72e4493cac63b3e69
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name nginx1 -p 8080:80 -v html
_volume:/usr/share/nginx/html -v nginx_config_volume:/etc/nginx nginx
c2f335ca74517b13f5e6aa9dca02ecd666fb1c21aad2c9ab2f7bd6a5fc4396f5
```

localhost:8080

hello albraa from docker Nginx

Problem 2:

Run a container Nginx with name nginx-bind-mount and attach 2 volumes

using bind mount under any paths

```
albraa@albraa-Cyborg-15-A13UDX:~$ mkdir -p ~/nginx-html
mkdir -p ~/nginx-config
albraa@albraa-Cyborg-15-A13UDX:~$ echo "<h1>Hello from Bind Mount Nginx</h1>" >
~/nginx-html/index.html
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name nginx-bind-mount -p 8010:
80 -v ~/nginx-html:/usr/share/nginx/html nginx
b4b40e8afa2f48d56db2773b5816f1e2eaae12eca554676c37c8b9f24bfed7c0
albraa@albraa-Cyborg-15-A13UDX:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS
   PORTS                                       NAMES
b4b40e8afa2f   nginx     "/docker-entrypoint.…"   9 seconds ago    Up 8 seconds
   0.0.0.0:8010->80/tcp, [::]:8010->80/tcp   nginx-bind-mount
c2f335ca7451   nginx     "/docker-entrypoint.…"   33 minutes ago   Up 33 minutes
   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nginx1
ea60b2cb873d   nginx     "/docker-entrypoint.…"   35 minutes ago   Up 35 minutes
   80/tcp                                      nginx2
```

Remove the container

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker rm -f nginx-bind-mount
nginx-bind-mount
```

• Run a new container with the following:

o Attach the two volumes that were attached to the previous container

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker run -d --name nginx-bind-mount-new -p 8
020:80 -v ~/nginx-html:/usr/share/nginx/html -v ~/nginx-config:/etc/nginx/conf.d
 nginx
db90f6a9225b16f0ba649b53f3b4b26579d673192145333640e3bcf4aaa1250d
```

Check the old data in the new containers

```
albraa@albraa-Cyborg-15-A13UDX:~$ docker exec -it nginx-bind-mount-new cat /usr/
share/nginx/html/index.html
<h1>Hello from Bind Mount Nginx</h1>
albraa@albraa-Cyborg-15-A13UDX:~$
```

localhost:8020

Hello from Bind Mount Nginx